Locality and unitarity from hidden zeros in $\mathrm{tr}(\phi^3)$: a direct computation at four and five points

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April 17, 2026

Abstract

We give an elementary, self-contained proof that locality and unitarity of the color-ordered tree amplitude of $\mathrm{tr}(\phi^3)$ emerge – rather than being assumed – from a short list of *hidden zeros* in planar kinematic space. Starting from the most general rational ansatz with the correct scaling, we demand only that it vanish on the hidden-zero loci. The computation is direct and arithmetic: at four points one linear locus forces the two Feynman-channel coefficients to be equal; at five points, five cyclically related loci kill every "wrong" term in a 15-parameter ansatz and equate all nearest-neighbor coefficients. The surviving object is the planar cubic amplitude, with the correct pole structure and residues.

1 Setup

Label the external legs of a color-ordered n-point amplitude of $\mathrm{tr}(\phi^3)$ cyclically by $1, 2, \ldots, n$. The *planar Mandelstam variables*

$$X_{ij} \;=\; (p_i + p_{i+1} + \cdots + p_{j-1})^2, \qquad 1 \le i < j \le n,\ (i,j) \text{ non-adjacent},$$

are in bijection with the *chords* of the n-gon $12\cdots n$, and there are $\frac{n(n-3)}{2}$ of them: two for $n = 4$, five for $n = 5$, nine for $n = 6$.

A *triangulation* is a maximal set of non-crossing chords; it has $n - 3$ elements and carves the n-gon into $n - 2$ triangles. Triangulations are in bijection with planar cubic Feynman diagrams: a chord is a propagator $1/X$, a triangle is a cubic vertex. The number of triangulations is the Catalan number C_{n-2}: two for $n = 4$, five for $n = 5$, fourteen for $n = 6$.

The tree amplitude is the sum over triangulations of the product of propagators,

$$A_n \;=\; \sum_{T \text{ triangulation}} \prod_{c \in T} \frac{1}{X_c}. \tag{1}$$

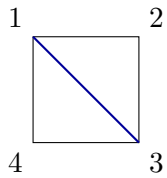
The question. Forget the Lagrangian. Write down the most general rational function B of the chord variables with the scaling that a cubic amplitude must have. Then demand that B vanish on the *hidden-zero loci* – certain linear subspaces of kinematic space on which A_n is known to vanish. Is that enough to recover A_n?

For $n = 4$ and $n = 5$: yes, up to one overall constant. Locality (simple poles only, on planar channels) and unitarity (the correct residue on every pole) are outputs, not inputs.

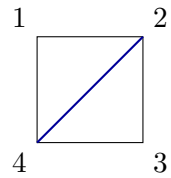
2 Warm-up: four points

The two triangulations of the square

The four-point polygon is the square with cyclic labels $1, 2, 3, 4$. It has two chords, $(1, 3)$ and $(2, 4)$, and each by itself is already a triangulation ($n - 3 = 1$), giving the two planar Feynman diagrams:



chord $(1, 3)$: $\dfrac{1}{X_{13}} = \dfrac{1}{s}$



chord $(2, 4)$: $\dfrac{1}{X_{24}} = \dfrac{1}{t}$

Write $x_1 = X_{13} = s$ and $x_2 = X_{24} = t$. The planar tree amplitude is

$$A_4 \;=\; \frac{1}{x_1} + \frac{1}{x_2} \;=\; \frac{1}{s} + \frac{1}{t}.$$

Ansatz

The amplitude has mass dimension $-2(n - 3) = -2$. The simplest Laurent ansatz with the correct scaling, built from the two chord variables and containing only simple planar poles, is

$$B_4 \;=\; \frac{b_1}{x_1} + \frac{b_2}{x_2}. \tag{2}$$

A double-pole term $1/x_i^2$ has the wrong dimension (-4), and a mixed product $1/(x_1 x_2)$ likewise has dimension -4; including either would require a numerator factor that reintroduces a non-planar singularity. In short: at four points dimensional analysis alone lets us start with (2) and nothing more.[1]

The four-point hidden zero

For four massless external legs, momentum conservation gives $s + t + u = 0$, where $u = X_{14}$ is the non-planar channel. The four-point hidden zero is the locus $u = 0$, i.e.

$$\mathcal{L}_4 : \qquad x_2 \;=\; -\,x_1. \tag{3}$$

On $\mathcal{L}_4$ the planar amplitude evaluates to $1/x_1 + 1/(-x_1) = 0$. Our ansatz gives

$$B_4\big|_{\mathcal{L}_4} \;=\; \frac{b_1}{x_1} + \frac{b_2}{-x_1} \;=\; \frac{b_1 - b_2}{x_1}.$$

Vanishing for every x_1 along $\mathcal{L}_4$ forces

$$\boxed{b_1 \;=\; b_2 \;=:\; c.} \tag{4}$$

[1] The parallel with the five-point ansatz (7) below, which *does* carry double poles, is instructive. At five points the dimension -4 forces products of two inverse chord variables; we then get to watch the hidden zeros explicitly kill the double-pole terms. At four points that work is done for us before we begin.

Hence

$$B_4 \;=\; c\left(\frac{1}{x_1}+\frac{1}{x_2}\right) \;=\; c\,A_4.$$

The single-pole structure and the equal weight on the two Feynman channels are both *derived*; we put neither in.

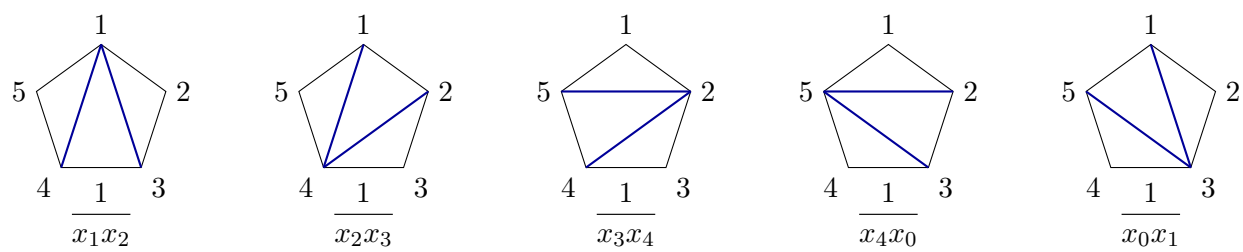
3 Five points

The five triangulations of the pentagon

Label the vertices of the pentagon $1,2,3,4,5$ cyclically. The five chords are

$$(1,3),\;(1,4),\;(2,4),\;(2,5),\;(3,5).$$

A triangulation is a pair of non-crossing chords, and in the pentagon two chords are non-crossing iff they share a leg. Each external leg lies on exactly one such pair, giving the five triangulations $C_3 = 5$:



The label beneath each diagram is the product of propagators that the triangulation contributes to the amplitude.

Cyclic labeling of the chords

For the computation below it is useful to relabel the chords by a single cyclic index, following the order in which they appear as we walk around the pentagon of chords:

$$x_1 \;=\; X_{13},\quad x_2 \;=\; X_{14},\quad x_3 \;=\; X_{24},\quad x_4 \;=\; X_{25},\quad x_0 \;=\; X_{35}. \tag{5}$$

Two chords share a leg iff their cyclic indices differ by ± 1 mod 5. Two chords cross iff their indices differ by ± 2 mod 5. The five triangulations are therefore in bijection with the five cyclically adjacent pairs (x_i, x_{i+1}), and

$$A_5 \;=\; \frac{1}{x_0x_1}+\frac{1}{x_1x_2}+\frac{1}{x_2x_3}+\frac{1}{x_3x_4}+\frac{1}{x_4x_0} \;=\; \sum_{i\in\mathbb{Z}/5}\frac{1}{x_i\,x_{i+1}}. \tag{6}$$

Ansatz

The five-point amplitude has mass dimension $-2(n-3) = -4$, so each monomial in a Laurent ansatz is a product of *two* inverse chord variables. The most general such function is

$$B \;=\; \sum_{0\le i\le j\le 4}\frac{a_{ij}}{x_i\,x_j}, \tag{7}$$

with 15 free coefficients: five diagonal a_{ii}/x_i^2 (double poles) and ten off-diagonal $a_{ij}/(x_ix_j)$. We do *not* assume simple poles.

The five hidden-zero loci

The pentagon has a $\mathbb{Z}/5$ symmetry acting on the chord labels (5). There are correspondingly five hidden-zero loci, one per vertex $k \in \{0,1,2,3,4\}$ of the cyclic chord diagram. Each locus imposes two linear constraints on the chord variables, leaving three free parameters, one of which is x_k itself:

k	linear constraints		free vertex
0	$x_4 = -x_1$	$x_3 = x_2 - x_1$	x_0
1	$x_0 = -x_2$	$x_4 = x_3 - x_2$	x_1
2	$x_1 = -x_3$	$x_0 = x_4 - x_3$	x_2
3	$x_2 = -x_4$	$x_1 = x_0 - x_4$	x_3
4	$x_3 = -x_0$	$x_2 = x_1 - x_0$	x_4

The claim (verifiable in a line, below) is that A_5 from (6) vanishes identically on each $\mathcal{L}_k$. We will demand the same of B and watch the coefficients collapse.

Proposition 1. $A_5\big|_{\mathcal{L}_k} = 0$ *for* $k = 0,1,2,3,4$.

Proof. By cyclic symmetry it suffices to check $k = 0$. Substituting $x_4 = -x_1,\ x_3 = x_2 - x_1$ into (6),

$$A_5\big|_{\mathcal{L}_0} = \underbrace{\frac{1}{x_0 x_1} + \frac{1}{(-x_1)x_0}}_{=0} + \frac{1}{x_1 x_2} + \frac{1}{x_2(x_2 - x_1)} + \frac{1}{(x_2 - x_1)(-x_1)}.$$

The remaining sum combines over the common denominator $x_1 x_2 (x_2 - x_1)$ to $\big[(x_2 - x_1) + x_1 - x_2\big]/\big[x_1 x_2 (x_2 - x_1)\big] = 0$. □

Case $k = 0$ in detail

Fix $\mathcal{L}_0$: $x_4 = -x_1,\ x_3 = x_2 - x_1$, with x_0, x_1, x_2 independent. Organize (7) by its dependence on x_0:

$$B\big|_{\mathcal{L}_0} = \underbrace{\frac{a_{00}}{x_0^2}}_{\mathcal{O}(x_0^{-2})} + \underbrace{\frac{1}{x_0}\left(\frac{a_{01}}{x_1} + \frac{a_{02}}{x_2} + \frac{a_{03}}{x_3} + \frac{a_{04}}{x_4}\right)}_{\mathcal{O}(x_0^{-1})} + \underbrace{\sum_{1 \le i \le j \le 4} \frac{a_{ij}}{x_i x_j}}_{\mathcal{O}(x_0^{0})}. \tag{8}$$

Since x_0 is independent, each bracketed piece must vanish on its own.

(a) Order x_0^{-2}. $a_{00}/x_0^2 = 0$ for all x_0 gives

$$\boxed{a_{00} = 0.}$$

(b) Order x_0^{-1}. The bracket in the middle of (8) must vanish as a function of x_1, x_2. With $x_3 = x_2 - x_1$ and $x_4 = -x_1$,

$$\frac{a_{01}}{x_1} + \frac{a_{02}}{x_2} + \frac{a_{03}}{x_2 - x_1} + \frac{a_{04}}{-x_1} = 0 \qquad \text{for all } x_1, x_2. \tag{9}$$

Two limits extract two coefficients for free.

Limit 1. Fix x_2; send $x_1 \to \infty$. Then $1/x_1 \to 0$, $1/(x_2 - x_1) \to 0$, $1/(-x_1) \to 0$, and only a_{02}/x_2 survives. Vanishing for all x_2 gives

$$\boxed{a_{02} = 0.}$$

Limit 2. Fix $x_3 = x_2 - x_1$; send $x_1 \to \infty$ (so $x_2 = x_1 + x_3 \to \infty$ too, and $x_4 = -x_1 \to -\infty$). Then $1/x_1 \to 0$, $1/x_2 \to 0$, $1/(-x_1) \to 0$, and only a_{03}/x_3 survives. Vanishing for all x_3 gives

$$\boxed{a_{03} = 0.}$$

What is left of (9) is

$$\frac{a_{01}}{x_1} + \frac{a_{04}}{-x_1} = \frac{a_{01} - a_{04}}{x_1} = 0 \quad \forall x_1 \quad \Longrightarrow \quad \boxed{a_{01} = a_{04}.}$$

(c) Order x_0^0. The third bracket of (8) depends only on x_1, x_2, x_3, x_4. It is handled automatically by the analogous analysis on the other four loci: Case $\mathcal{L}_k$ extracts the a_{ij} with $k \in \{i, j\}$ from the order x_k^{-2} and x_k^{-1} pieces. The $\mathcal{O}(x_0^0)$ piece at Case 0 therefore carries only information already constrained by Cases $1, 2, 3, 4$, and – as we verify at the end of the section – vanishes identically once those constraints are imposed.

Collecting all five cases

By cyclic symmetry Case k produces, with indices read mod 5,

$$a_{kk} = 0, \qquad a_{k,k+2} = a_{k,k+3} = 0, \qquad a_{k,k-1} = a_{k,k+1}. \tag{10}$$

Running $k = 0, 1, 2, 3, 4$ and remembering $a_{ij} = a_{ji}$:

(i) *Diagonal.* All five $a_{ii} = 0$.

(ii) *Skip-one off-diagonal.* The five pairs (i, j) with $|i-j| \equiv 2 \pmod 5$ – that is, $(0, 2), (1, 3), (2, 4), (0, 3), (1, 4)$ – all satisfy $a_{ij} = 0$.

(iii) *Nearest-neighbor.* The five equalities $a_{01} = a_{04}$, $a_{10} = a_{12}$, $a_{21} = a_{23}$, $a_{32} = a_{34}$, $a_{43} = a_{40}$ – combined with symmetry – chain together to a single scalar:

$$a_{01} = a_{12} = a_{23} = a_{34} = a_{40} =: c.$$

Substituting back into (7),

$$B = c\left(\frac{1}{x_0x_1} + \frac{1}{x_1x_2} + \frac{1}{x_2x_3} + \frac{1}{x_3x_4} + \frac{1}{x_4x_0}\right) = c\,A_5. \tag{11}$$

Locality (no $1/x_i^2$; no non-adjacent cross-products) and unitarity (equal weight on every planar channel, matching the residue of lower-point amplitudes on every simple pole $x_k \to 0$) are both outputs.

Consistency: the $\mathcal{O}(x_0^0)$ piece closes

To tie off Case 0: once (i)–(iii) are imposed, the third bracket of (8) reduces to

$$c\left(\frac{1}{x_1x_2} + \frac{1}{x_2x_3} + \frac{1}{x_3x_4}\right) = c\left(\frac{1}{x_1x_2} + \frac{1}{x_2(x_2 - x_1)} - \frac{1}{x_1(x_2 - x_1)}\right),$$

and, just as in the proof of Proposition 1, the bracket combines to 0. Case 0 is self-consistent; by cyclic symmetry so are Cases $1, 2, 3, 4$. $\square$

What we actually did

At four points a single linear locus, $u = 0$, collapses a two-parameter family to a one-parameter family, and the surviving parameter is the overall normalization of A_4.

At five points a 15-parameter family is collapsed by five linear loci, one per pentagon vertex, to a one-parameter family: the surviving parameter is the overall normalization of A_5. The hidden zeros do not just constrain the amplitude; they *carve it out* of the space of general rational functions.

The pattern extrapolates. At six points the ansatz has $\binom{9+1}{2} = 45$ coefficients, and one expects six cyclic hidden zeros together with longer-range factorization loci to reduce the count to one. The combinatorics is no longer a single page, but the logic is the same: demand vanishing on the right codimension-2 planes, and locality and unitarity come out for free.